Project 2: Enzyme kinetics

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Data

The data set contains 23 observations of enzyme reaction rates measured at varying substrate concentrations for two treatment groups - treated and untreated.

QUESTION 1: MODEL SPECIFICATION

```
model {
  # LIKELIHOOD
  for (i in 1:N) {
      Y[i] ~ dnorm(mu[i], tau)
      mu[i] <- (Vmax[state[i]] * x[i]) / (K[state[i]] + x[i])
  }

  # PRIORS
  for (j in 1:2) {
    Vmax[j] ~ dunif(0, 1000)
  }

  for (j in 1:2) {
    K[j] ~ dunif(0, 10)
  }

  tau ~ dgamma(0.001, 0.001)
  # variance
  sigma2 <- 1 / tau
  # standard deviation
  sigma <- 1 / sqrt(tau)
}
```

Likelihood. Each observed reaction rate Y_i is assumed to follow a Normal distribution:

$$Y_i \sim \text{Normal}(\mu_i, \tau)$$

It is centered on the true mean rate μ_i with precision τ that controls how much the observed rates scatter around the Michaelis-Menten curve.

Nonlinear mean structure. The mean μ_i follows the Michaelis-Menten equation:

$$\mu_i = \frac{V_{max,j} \cdot x_i}{K_j + x_i}$$

Rather than a straight line, this is a saturating curve - reaction rate rises steeply at low concentrations then levels off as the enzyme approaches its capacity. The index $j =$ `state[i]` ensures each observation uses the parameters of its own group, so the curve is fitted separately for treated and untreated.

Prior distributions. Uninformative priors were assigned to all parameters:

- $V_{max,j} \sim \text{Uniform}(0, 1000)$: must be positive as it is a rate. The upper bound of 1000 is far above the maximum observed rate of ~207, making this effectively uninformative.
- $K_j \sim \text{Uniform}(0, 10)$: must be positive as it is a concentration. The upper bound of 10 is well above the maximum observed concentration of 1.1, so again effectively uninformative.
- $\tau \sim \text{Gamma}(0.001, 0.001)$: the standard vague prior for precision in JAGS. Near-zero shape and rate parameters spread mass thinly across a wide range of variances, expressing ignorance about the noise level in the data.

Role of each parameter:

- $V_{max,j}$ - the asymptote of the curve; the maximum reaction rate the enzyme can approach but never exceed. One value per group, capturing each group's capacity ceiling.
- K_j - the half-saturation constant; the concentration at which the reaction rate reaches exactly half of V_{max}. A small K means the enzyme saturates quickly and is efficient. One value per group.
- τ (and $\sigma = 1/\sqrt{\tau}$) - the precision and standard deviation of the observations around the curve. Shared across both groups, assuming equal variability in the measurements regardless of treatment.

QUESTION 2: RUN THE MCMC ALGORITHM

Number of chains. Three parallel chains were run, each starting from different initial values. Running multiple chains allows us to assess whether they converge to the same posterior distribution regardless of starting point.

Burn-in. The first 5,000 iterations were discarded. During this period the sampler is still moving toward the target distribution and samples are not yet representative of the posterior. A burn-in of 5,000 is conservative and appropriate given the nonlinear mean structure, which can slow mixing.

Iterations. Following burn-in, 10,000 iterations were retained per chain, yielding 30,000 posterior samples in total. No thinning was applied as autocorrelation diagnostics confirmed the chains were mixing well.

Convergence Diagnostics

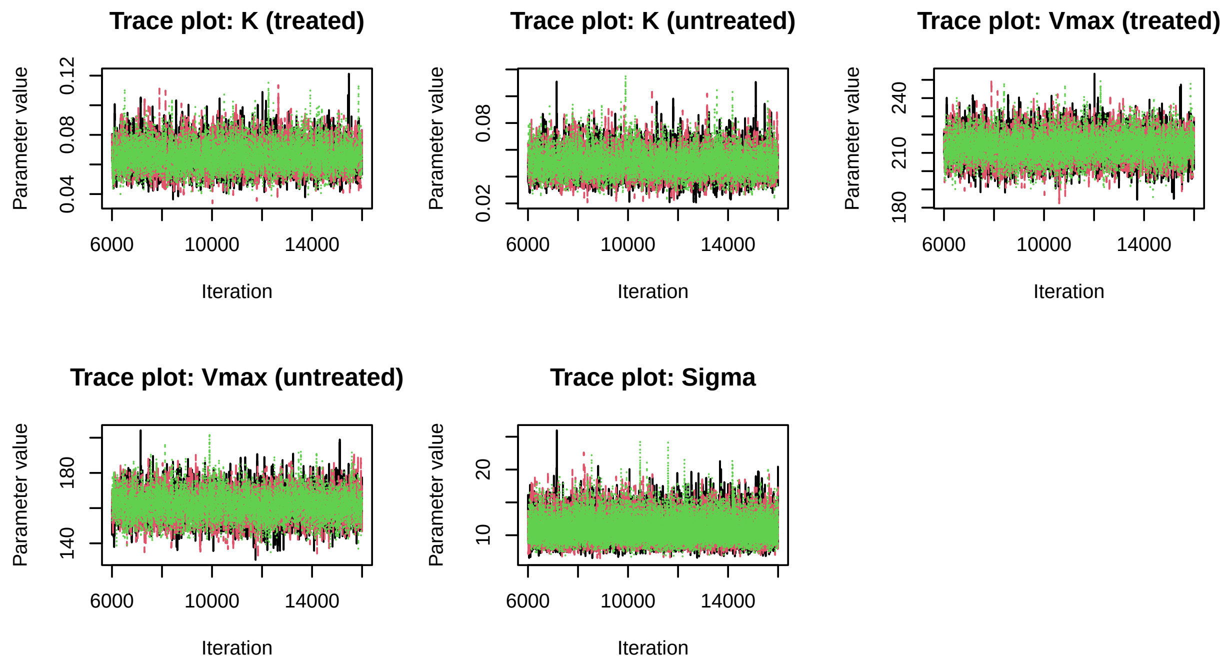


Table 1: Gelman-Rubin convergence diagnostics

	Parameter	Point Estimate	Upper C.I
K[1]	K (treated)	1	1
K[2]	K (untreated)	1	1
Vmax[1]	Vmax (treated)	1	1
Vmax[2]	Vmax (untreated)	1	1
sigma	Sigma	1	1

Trace plots. The trace plots show all three chains overlapping completely throughout the sampling period, with no visible trends, drifts, or periods where a chain becomes stuck. This stable, well-mixed behavior confirms the chains are sampling from the same stationary distribution.

Gelman-Rubin statistic ($\hat{R}$). The $\hat{R}$ statistic compares within-chain variance to between-chain variance. A value of 1 indicates perfect convergence; values below 1.1 are considered acceptable. All parameters returned $\hat{R} = 1$, well below the 1.1 threshold, confirming the chains have converged to the same target distribution.

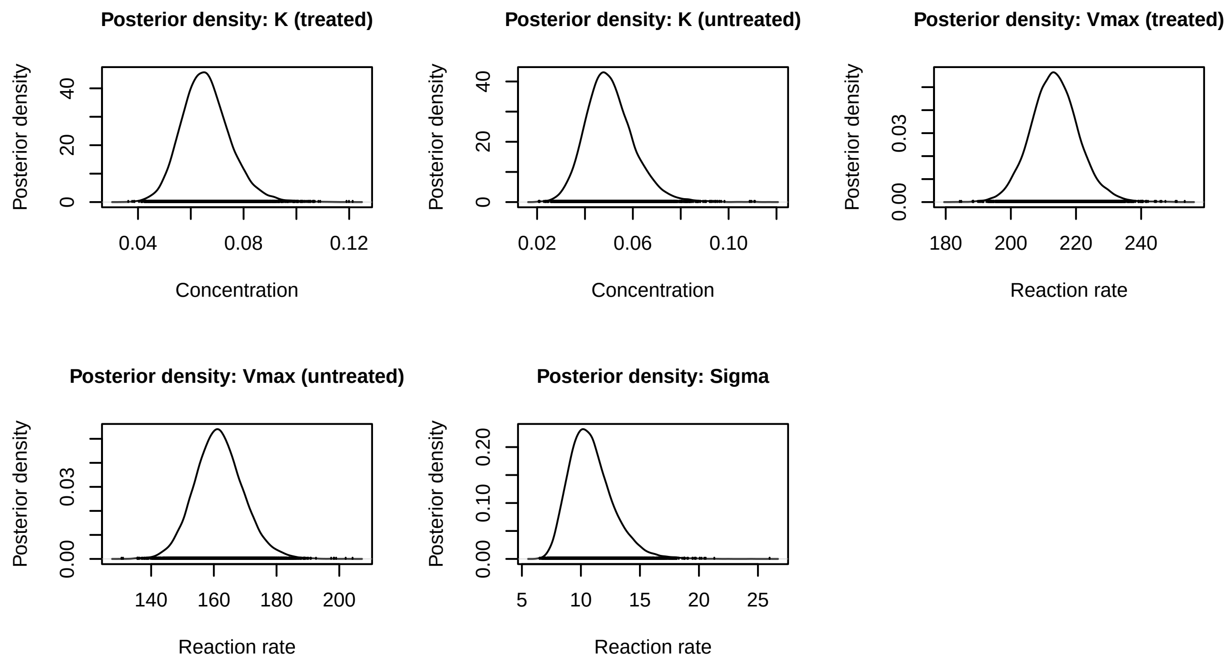
Conclusion on Convergence Both diagnostics agree: the chains have converged. The $\hat{R}$ values are all below 1.1 and the trace plots show stable, well-mixed chains throughout. The 30,000 posterior samples retained after burn-in provide a reliable basis for posterior inference on V_{max}, K, and σ.

QUESTION 3: POSTERIOR SUMMARIES

Table 2: Posterior summaries for all model parameters

	Point estimates		95% Credible interval		
	Mean	SD	2.5%	Median	97.5%
K (treated)	0.0658	0.0093	0.0495	0.0652	0.0859
K (untreated)	0.0504	0.0101	0.0331	0.0495	0.0729

Vmax (treated)	213.5123	7.4711	199.2667	213.3175	229.1575
Vmax (untreated)	161.6845	7.7942	147.0464	161.4231	177.9990
Sigma	10.8691	1.8815	7.9324	10.6318	15.1914



V_{max} - Maximum Reaction Rate. The posterior mean for the treated group is 213.5 (95% CI: 199.3, 229.2) and for the untreated group is 161.7 (95% CI: 147.0, 178.0). The credible intervals do not overlap, providing strong evidence that treatment increases the enzyme's maximum reaction capacity.

K - Half-Saturation Constant. The posterior mean for the treated group is 0.066 (95% CI: 0.050, 0.090) and for the untreated group is 0.050 (95% CI: 0.033, 0.073). The overlapping credible intervals suggest the difference in substrate affinity between groups is less certain than the difference in Vmax.

σ - Observation Noise. The posterior mean of the residual standard deviation is 10.87 (95% CI: 7.93, 15.19). This describes the common scatter of observations around the fitted Michaelis-Menten curves for both groups, consistent with moderate measurement variability in the enzyme reaction.

Overall. The treatment raises the enzyme's maximum reaction rate without meaningfully altering its substrate affinity. The credible intervals on V_{max} are completely separated while those on K overlap considerably, telling two very different stories about what the treatment does and does not affect.

QUESTION 4: POSTERIOR RATIOS

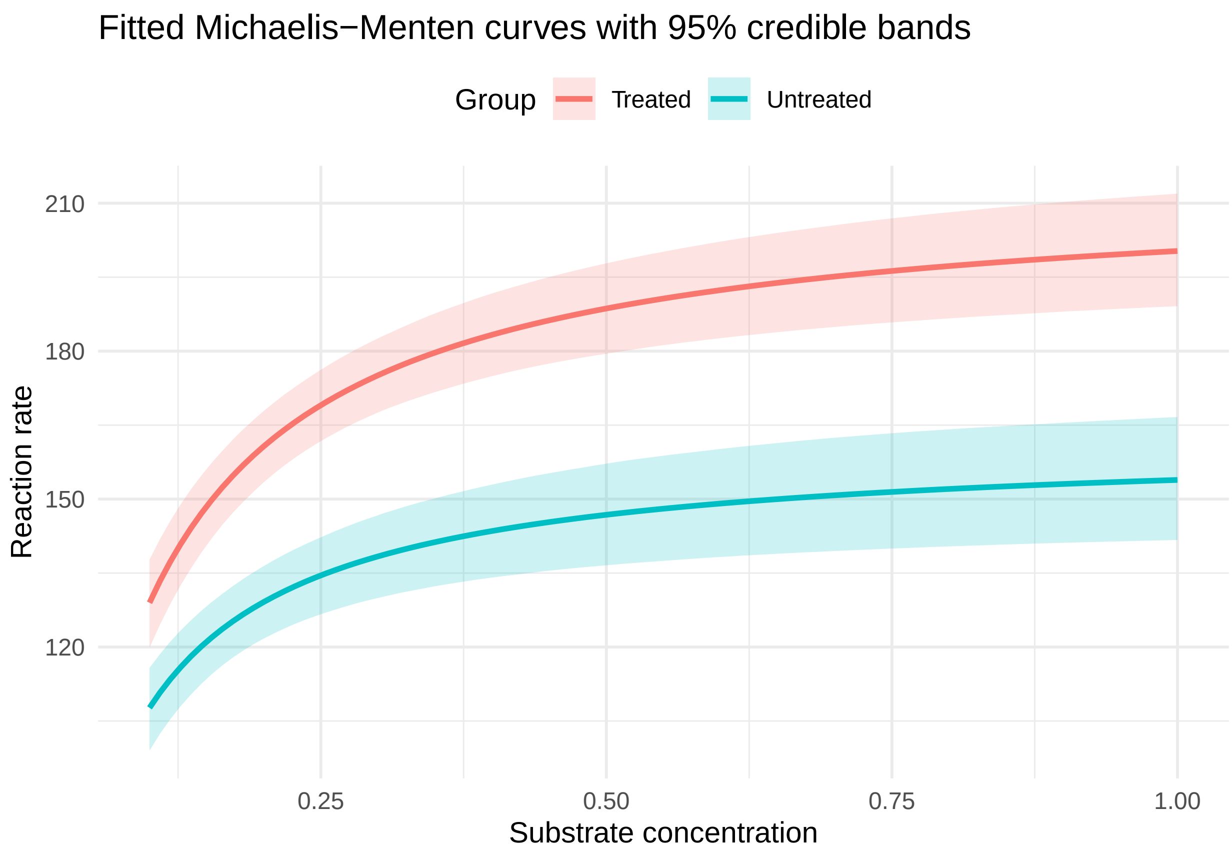
Table 3: Posterior summaries for parameter ratios

	Point estimates		95% Credible interval		
	Mean	SD	2.5%	Median	97.5%
Vmax (treated / untreated)	1.3236	0.0789	1.1737	1.3211	1.4869
K (treated / untreated)	1.3574	0.3393	0.8189	1.3138	2.1533

V_{max} Ratio. The posterior mean is approximately 1.32 (95% CI: 1.17, 1.49). The entire credible interval lies above 1, confirming the treated enzyme's maximum reaction rate is credibly higher by approximately 32%.

K Ratio. The posterior mean is approximately 1.36 (95% CI: 0.82, 2.15). The credible interval includes 1, indicating no credible difference in substrate affinity between groups.

QUESTION 5: FITTED REACTION CURVES



The fitted curves are obtained by plugging every posterior sample of V_{max} and K into the Michaelis-Menten equation across a fine concentration grid from 0.1 to 1, yielding 30,000 plausible curves per group. Parameter uncertainty is therefore fully propagated into the uncertainty bands.

Shape. Both groups follow the expected saturating curve. Observed data points fall comfortably within the credible bands, confirming a good model fit.

Separation. The treated curve sits clearly and consistently above the untreated curve across the entire concentration range with no crossing, reflecting the higher V_{max} in the treated group.

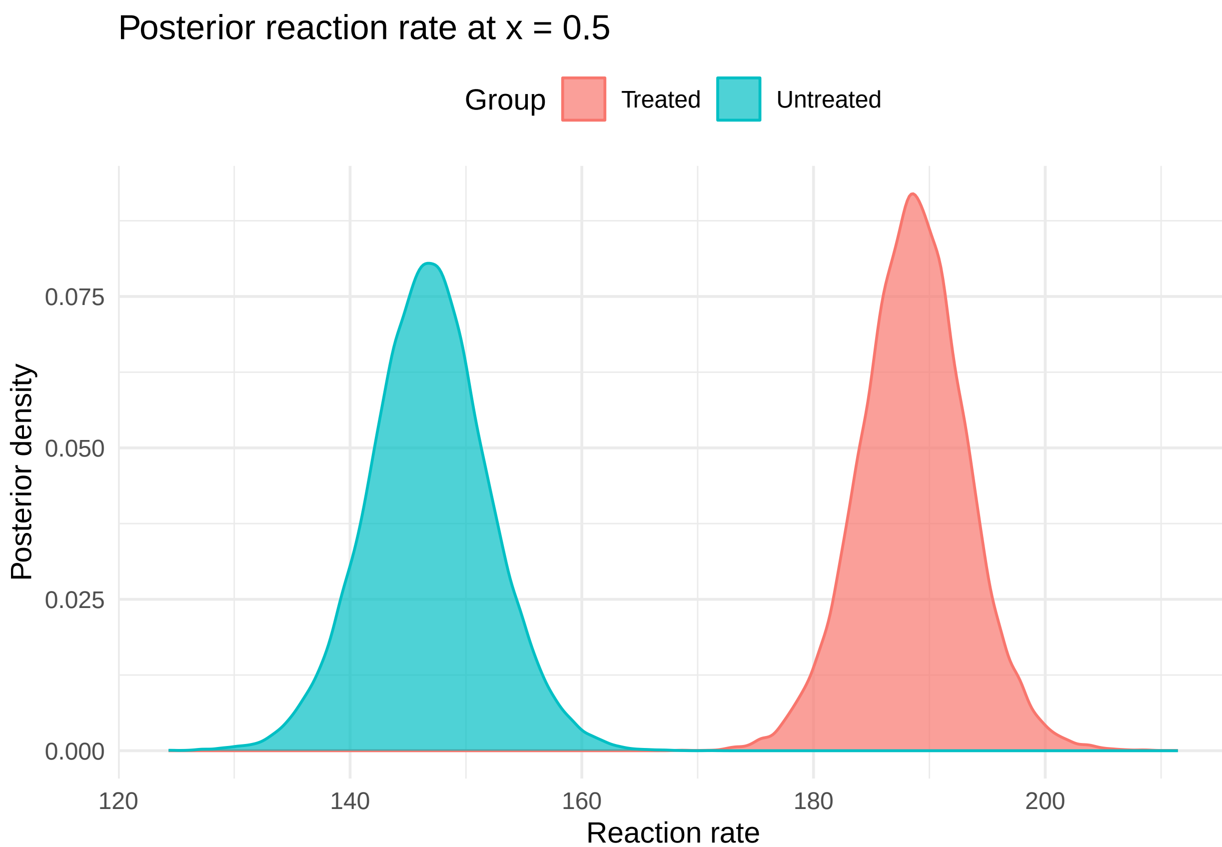
Uncertainty bands. The bands are narrow for both groups and widen slightly at higher concentrations where uncertainty in V_{max} has the greatest influence.

Similar shape, different ceiling. Because K does not credibly differ between groups, the curves share the same general curvature. The treatment shifts the curve upward rather than changing its shape.

QUESTION 6: POSTERIOR DISTRIBUTION AT CONCENTRATION OF 0.5

Table 4: Posterior reaction rate at concentration x = 0.5

	Point estimates		95% Credible interval		
	Mean	SD	2.5%	Median	97.5%
Rate at x=0.5 (treated)	188.6473	4.5794	179.4720	188.6432	197.8009
Rate at x=0.5 (untreated)	146.8162	5.1656	136.5804	146.7994	157.1816



At $x = 0.5$, the posterior mean reaction rate for the treated group is approximately 188.6 (95% CI: 179.5, 197.8) and for the untreated group approximately 146.8 (95% CI: 136.6, 157.2). The two posterior distributions are clearly separated with non-overlapping credible intervals, confirming the treatment effect is real and meaningful at this concentration. Both distributions are smooth and unimodal, reflecting well-identified parameters and good chain mixing. The relatively narrow distributions reflect the fact that $x = 0.5$ lies well within the observed concentration range, where the curve is well-constrained.

QUESTION 7: POSTERIOR PROBABILITIES

```
P(Vmax treated > Vmax untreated | data) = 1
 P(K treated < K untreated | data) = 0.1213
```

The estimates were obtained by using Monte Carlo proportions over the posterior distribution.

Interpretation:

Since $P(Vmax\_treated > Vmax\_untreated \mid data)$ is 1 or very close to it, we have strong evidence, that treatment with Puromycin increases the maximum reaction rate of the enzyme.

$P(K\_treated < K\_untreated \mid data) \approx 0.12$ can be reversed so that $P(K\_treated > K\_untreated \mid data) \approx 0.88$, which means that there is strong evidence, that the treated enzyme requires higher substrate concentration to reach half its maximum rate.

These results imply that Puromycin increases maximum rate, however it also increases the concentration needed in order to reach that rate.

QUESTION 8: POSTERIOR DISTRIBUTION WHEN CONCENTRATION AT 80% Vmax

To get the x (concentration) where our expected reaction reaches 80% of the maximum, we worked out x analytically.

$$\frac{V_{max} \cdot x}{K + x} = 0.8V_{max}$$

$$\frac{x}{K + x} = 0.8$$

$$x = 0.8K + 0.8x$$

$$0.2x = 0.8K$$

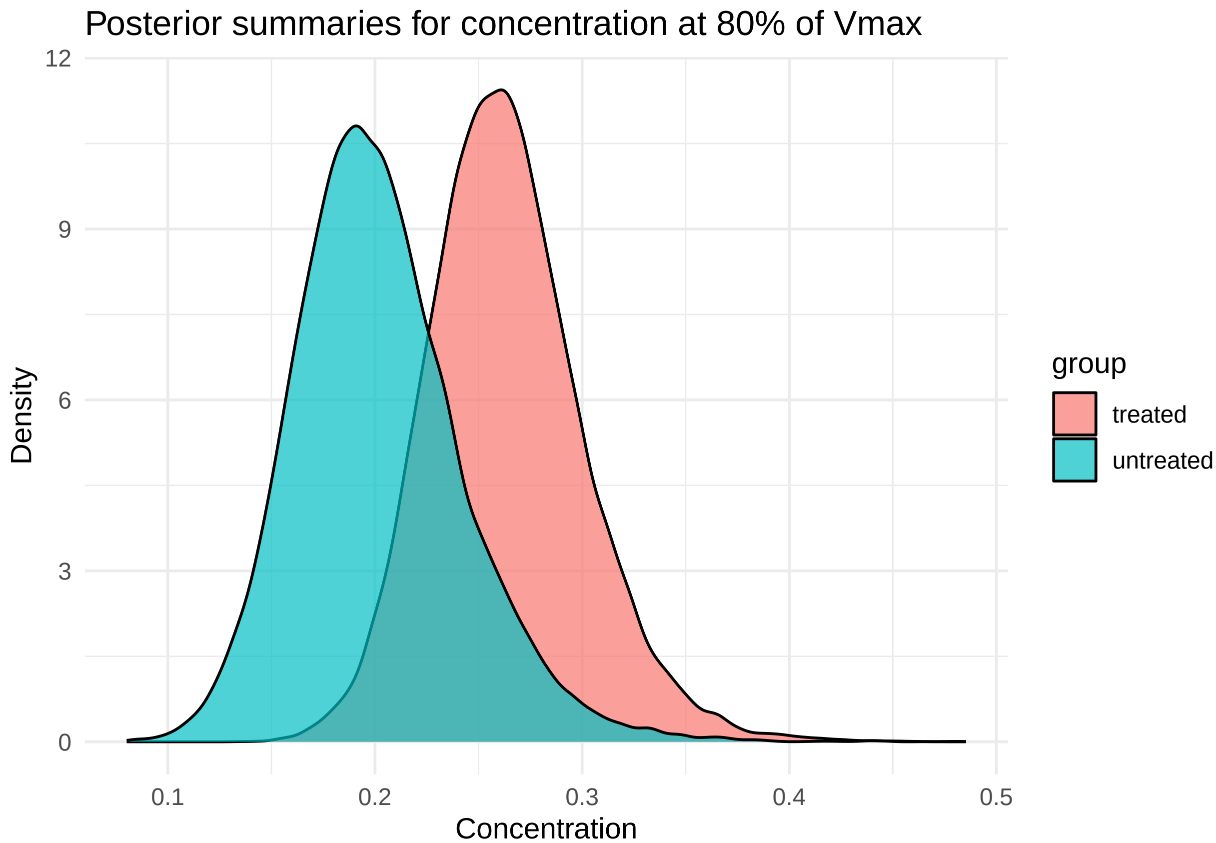
$$x = 4K$$

This means that we need 4 times the half-saturation concentration to reach 80% of the maximum rate.

Since we have a posterior distribution of K, we can transform it using the previously derived formula $x = 4K$ to get the posteriors for the concentrations at 80% V_max

Table 5: Posterior summaries for concentration at 80% of Vmax

	Point estimates		95% Credible interval		
Group	Mean	SD	2.5%	Median	97.5%
Treated	0.2632	0.0371	0.1978	0.2610	0.3437
Untreated	0.2017	0.0404	0.1323	0.1981	0.2916



Interpretation: The posterior distributions are well separated, only partially overlapping. We can see from the table that the posterior mean concentration required to reach 80% of Vmax is 0.263 ppm (95% CI: 0.198, 0.344) for the treated group and 0.202 ppm (95% CI: 0.132, 0.292) for the untreated group.

This means that the untreated group reaches 80% of its maximum rate with lower concentration than the treated enzyme, though the overlapping credible intervals mean this difference is not conclusive.

All code used in this analysis is provided in the accompanying R script `bayes_project.R`. This report presents outputs and interpretations only.